Zhang Zhenhui

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Gender: Male | Native place: Fuzhou, Fujian

EDUCATION

Xidian University Sep 2019–Jul 2022

Space Engineering | M.E. | Space Science and Technology College Xi'an City

GPA: 3.60

Civil Aviation University of China Sep 2015–Aug 2019

Communication Engineering | B.E. | Electronic Information and Automation College Tianjin City

GPA: 3.46

HONORARY AWARDS

Best Idea /Best Hands on Ability Award of Honeywell-Star Jul 2017/Jul 2016

First Prize of College Students' Engineering Training Comprehensive Ability Competition May 2017

First/Third Prize of National Undergraduate Electronic Design Competition in Tianjin Dec 2017/Dec 2016

"Beidou Cup" Science and Technology Competition National Second/Excellent Prize May 2017/May 2016

Second Prize of NXP National College Students' Smart Car Competition in North China Aug 2017

First Prize of Tianjin Robot Competition Oct 2016

Second/Third Prize of Robot Competition in North China Nov 2017/ Nov 2016

PROJECT EXPERIENCE

Main control unit of scientific electronic control SPU Dec 2019-Up to Now

-- Shanghai Institute of Optics and precision machinery, Chinese Academy of Sciences

-- Head of Hardware Design

Project Introduction: the equipment will be used in the ultracold atomic physics experiment cabinet in the experimental module II "Mengtian" of China space station. By taking advantage of the microgravity environment, the equipment will create new extreme conditions, obtain the ultra-low temperature of 10-12K (PK) level that cannot be achieved on the ground, and build a super low temperature, large-scale and high-quality laboratory, An open experimental system of Bose and Fermi quantum degenerate working matter suitable for long time precise measurement. The controller is the main control module in the experimental system, which mainly realizes the functions of sending control commands to the laser subsystem and physical control unit, as well as the acquisition and processing of experimental data and images. Several boards are interconnected by VPX structure, and use several zynq ultrascale series FPGAs of Xilinx company as processing and control equipment. The configuration and control instructions required by the laser subsystem and physical control unit are transmitted through the network, and the experimental images are collected by usb-ccd camera, converted into HDMI signals and output to the display screen. Through the coordination of high-speed ADC, DAC and DDS devices, the laser intensity and frequency are monitored by AOM devices, and then the laser intensity and frequency are adjusted and stabilized by adjusting the output of DAC and DDS devices.

Main Tasks: as the project Hardware Director, complete the schematic drawing of all boards and PCB layout and wiring, including FPGA power supply circuit, PHY circuit, USB circuit; The hardware logic development of zynq device is completed. The user-defined IP based on Axi bus is written in Verilog language, and the axi-dma interconnection IP is added. The high-precision parallel work of multi-channel USB, network port, HDMI display, ADC, DAC, TTL, UART and SPI is realized. In addition, according to the special requirements of the space project, the circuit design and code writing are processed and verified.

Ultra high speed data acquisition and continuous storage system Jun 2019-May 2020

-- Xi'an Branch of China Academy of space technology (504 Institute)

-- FPGA Hardware Development Engineer

Project Description: the system uses two Xilinx virtex7 series FPGAs as the platform, uses two High-Speed ADCs with 5gsps sampling rate and 8bit data bit width for analog signal sampling, and transmits 80gbps sampling data to the computer memory for back-end processing in real time through PCIe.

Main work: as a member of the project team, completed the hardware logic development. Two vc709 development boards and four FMC sub boards of DSP 5gsps ADC are used for development and debugging. For a single FPGA development board, idelayer and iserders are used for high-speed data synchronization, and clock edge alignment algorithm is designed to eliminate metastable state; The multi-channel data automatic alignment algorithm is designed for the uneven data; The high-speed LVDS data received by FPGA is recombined with the idea of data splicing; The offset error, gain error and clock phase error of each ADC are eliminated to ensure equal interval sampling; Using MIG IP core to realize the control design of high-speed DDR3 chip, to meet the requirements of high-speed data storage and reading; The asynchronous problem of high speed parallel data transmission across clock domain is eliminated; With the cooperation of members in the same group, Xilinx's DMA for pcieip core is used for the development of high-speed data transmission of pciegen3 to realize 40Gbps stable data transmission and data dropping. Based on the principle of parallel alternating sampling, the clock with different phases is input to the two boards, and the data collected by the two boards are synchronized in the computer to realize the 80gbps data acquisition of the whole system.

A320-IPT Simulator Equipment Sep 2016-Sep 2019

-- Civil Aviation University of China

-- Project Leader

Project Description: During the University, participated in the research and development of QA320-IPT simulator equipment, as the project leader of the national project "building a simple aircraft simulator platform based on FSUIPC", and published a paper. So far, we have completed the design and processing of saddle frame, A320 throttle imitation, and combined with FSUIPC protocol to realize the docking with P3D flight simulation software.

PROFESSIONAL SKILLS

-- Proficient in using vivado software and Verilog hardware description language to develop FPGA devices, master the development process of FPGA and zynq devices of Xilinx company, and have rich experience in FPGA device design and development;

-- Proficient in the schematic design of various devices, able to independently carry out Multisim circuit simulation and ad circuit board design and drawing, with rich experience in circuit board design and development;

-- Familiar with the basic operation of Linux operating system and the development and design process of petalinux system, master the embedded development of 51 series, ardrino, STM32 Series MCU;

-- Proficient in using SW, UG and other three-dimensional modeling software, draftsman level 4, able to carry out simple structural design and assembly work;

-- Understand the use of Visual Studio software and the preliminary development of C #.

PERSONAL SUMMARY

-- Solid foundation, strong thirst for knowledge, team spirit, down-to-earth and serious work, strong sense of responsibility;

-- Have rich project experience about FPGA, strong research ability, and can solve problems independently; Team spirit and innovation spirit;

-- Have the ability to write technical documents, good communication and cooperation ability, strong text synthesis and expression ability;